

Practice Questions for 2024-10-01-SensitivityDuality

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1 Questions

1. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
 - a. x_1 decreases, x_2 increases, ζ increases; $t \leq 1000$
 - b. x_1 increases, x_2 decreases, ζ increases; $t \leq 3000$
 - c. x_1 decreases, x_2 increases, ζ increases; $t \leq 3000$
 - d. x_1 increases, x_2 decreases, ζ decreases; $t \leq 1000$
 - e. x_1 decreases, x_2 increases, ζ remains unchanged; $t \leq 1000$
2. Using complementary slackness (Slide 3), verify the optimality of the solution $x_1^* = 25$, $x_2^* = 75$ for the primal problem in Slide 1. What are the optimal values of the dual variables y_1^* and y_2^* , and how do they relate to the optimal objective function value?
 - a. $y_1^* = 32.5$, $y_2^* = 0.75$; Strong duality holds, $\zeta^* = 6250 = 100y_1^* + 4000y_2^*$.
 - b. $y_1^* = 0$, $y_2^* = 0$; Strong duality does not hold.
 - c. $y_1^* = 25$, $y_2^* = 75$; Strong duality holds, $\zeta^* = 6250 = 100y_1^* + 4000y_2^*$.
 - d. $y_1^* = 40$, $y_2^* = 70$; Strong duality does not hold.
 - e. $y_1^* = 70$, $y_2^* = 40$; Strong duality holds, $\zeta^* = 6250 = 100y_1^* + 4000y_2^*$.
3. Explain the economic interpretation of the dual variables y_1^* and y_2^* obtained in the previous question. How are these values related to the sensitivity of the optimal objective function value to changes in the right-hand side of the constraints?
 - a. They represent shadow prices; a unit increase in the right-hand side of the first constraint increases ζ by y_1^* , and a unit increase in the right-hand side of the second constraint increases ζ by y_2^* .
 - b. They represent the optimal quantities of resources used; they are not related to sensitivity analysis.
 - c. They represent the marginal costs of the resources; a unit increase in the right-hand side of either constraint decreases ζ by the corresponding dual variable.
 - d. They represent the optimal production levels; they are not related to sensitivity analysis.
 - e. They represent the reduced costs; they are not related to sensitivity analysis.

4. Consider the matrix representation of the linear program in Slide 4. If we partition the variable vector X into basic (X_B) and non-basic (X_N) variables, what is the significance of the reduced cost vector $\bar{C} = (B^{-1}N)^T c_B - c_N$ in determining the optimality of the current solution?
 - a. If $\bar{C} \geq 0$, the current solution is optimal; otherwise, it is not.
 - b. If $\bar{C} \leq 0$, the current solution is optimal; otherwise, it is not.
 - c. If any element of $\bar{C}$ is zero, the current solution is optimal.
 - d. If any element of $\bar{C}$ is positive, the current solution is optimal.
 - e. The reduced cost vector is irrelevant to determining optimality.

5. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
 - a. x_1 decreases, x_2 increases, ζ increases; $t \leq 1000$
 - b. x_1 increases, x_2 decreases, ζ increases; $t \leq 1000$
 - c. x_1 decreases, x_2 increases, ζ increases; $t \leq 3000$
 - d. x_1 increases, x_2 decreases, ζ decreases; $t \leq 1000$
 - e. x_1 decreases, x_2 increases, ζ remains unchanged; $t \leq 1000$

6. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
 - a. x_1 decreases, x_2 increases, ζ increases; $t \leq 1000$
 - b. x_1 increases, x_2 decreases, ζ increases; $t \leq 4000$
 - c. x_1 decreases, x_2 increases, ζ increases; $t \leq 4000$
 - d. x_1 increases, x_2 decreases, ζ decreases; $t \leq 1000$
 - e. x_1 decreases, x_2 increases, ζ remains unchanged; $t \leq 1000$

7. Using complementary slackness (Slide 3), verify the optimality of the solution $x_1^* = 25$, $x_2^* = 75$ for the primal problem in Slide 1. What are the optimal values of the dual variables y_1^* and y_2^* , and how do they relate to the optimal objective function value?
 - a. $y_1^* = 32.5, y_2^* = 0.75; 100y_1^* + 4000y_2^* = 6250 = \zeta^*$
 - b. $y_1^* = 0, y_2^* = 1.4; 100y_1^* + 4000y_2^* = 5600 \neq \zeta^*$
 - c. $y_1^* = 40, y_2^* = 70; 100y_1^* + 4000y_2^* = 290000 \neq \zeta^*$
 - d. $y_1^* = 25, y_2^* = 75; 100y_1^* + 4000y_2^* = 302500 \neq \zeta^*$
 - e. $y_1^* = 30, y_2^* = 1; 100y_1^* + 4000y_2^* = 7000 \neq \zeta^*$

8. Explain the economic interpretation of the dual variables y_1^* and y_2^* obtained in the previous question. How are these values related to the sensitivity of the optimal objective function value to changes in the right-hand side of the constraints?

- a. y_1^* and y_2^* represent the marginal increase in profit per unit increase in the right-hand side of the first and second constraints, respectively. The sensitivity of ζ^* to changes in the right-hand side is directly proportional to these values.
 - b. y_1^* and y_2^* represent the shadow prices of resources 1 and 2, respectively. The sensitivity of ζ^* to changes in the right-hand side is given by the dual variables.
 - c. y_1^* and y_2^* represent the marginal cost of resources 1 and 2, respectively. The sensitivity of ζ^* to changes in the right-hand side is inversely proportional to these values.
 - d. y_1^* and y_2^* represent the optimal quantities of resources 1 and 2, respectively. The sensitivity of ζ^* to changes in the right-hand side is independent of these values.
 - e. y_1^* and y_2^* represent the slack variables in the dual problem. The sensitivity of ζ^* to changes in the right-hand side is unrelated to these values.
9. Consider the matrix representation of the linear program in Slide 4. If we partition the variable vector X into basic (X_B) and non-basic (X_N) variables, what is the significance of the reduced cost vector $\bar{C} = (B^{-1}N)^T c_B - c_N$ in determining the optimality of the current solution?
- a. If $\bar{C} = 0$, the current solution is optimal. If any element of $\bar{C}$ is positive, the solution is not optimal.
 - b. If $\bar{C} > 0$, the current solution is optimal. If any element of $\bar{C}$ is negative, the solution is not optimal.
 - c. If $\bar{C} < 0$, the current solution is optimal. If any element of $\bar{C}$ is positive, the solution is not optimal.
 - d. The reduced cost vector is irrelevant to determining optimality.
 - e. If any element of $\bar{C}$ is zero, the current solution is optimal.
10. a. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
- A. x_1 decreases, x_2 increases, ζ increases; $t \leq 1000$
 - B. x_1 increases, x_2 decreases, ζ increases; $t \leq 1000$
 - C. x_1 decreases, x_2 increases, ζ increases; $t \leq 3000$
 - D. x_1 increases, x_2 decreases, ζ decreases; $t \leq 1000$
 - E. x_1 decreases, x_2 increases, ζ remains unchanged; $t \leq 1000$
- b. Using complementary slackness (Slide 3), verify the optimality of the solution $x_1^* = 25$, $x_2^* = 75$ for the primal problem in Slide 1. What are the optimal values of the dual variables y_1^* and y_2^* , and how do they relate to the optimal objective function value?
- A. $y_1^* = 32.5$, $y_2^* = 0.75$; $100y_1^* + 4000y_2^* = 6250 = \zeta^*$
 - B. $y_1^* = 25$, $y_2^* = 75$; $100y_1^* + 4000y_2^* = 302500 \neq \zeta^*$
 - C. $y_1^* = 0$, $y_2^* = 0$; $100y_1^* + 4000y_2^* = 0 \neq \zeta^*$
 - D. $y_1^* = 40$, $y_2^* = 70$; $100y_1^* + 4000y_2^* = 300000 \neq \zeta^*$
 - E. $y_1^* = 70$, $y_2^* = 40$; $100y_1^* + 4000y_2^* = 190000 \neq \zeta^*$

2 Answers

1. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
 - a. From Slide 7, increasing the right-hand side of the second constraint by t leads to the updated optimal solution $x_1^* = 25 - \frac{t}{40}$ and $x_2^* = 75 + \frac{t}{40}$. Thus, x_1 decreases and x_2 increases. The objective function becomes $\zeta^* = 6250 + 0.75t$, indicating an increase in ζ . For feasibility, we require $x_1^* \geq 0$ and $x_2^* \geq 0$, which implies $25 - \frac{t}{40} \geq 0$ and $75 + \frac{t}{40} \geq 0$. Solving these inequalities gives $t \leq 1000$ and $t \geq -3000$. Therefore, the maximum allowable increase is $t \leq 1000$.
 - b. The direction of change in x_1 and x_2 is incorrect. Slide 7 shows that x_1 decreases and x_2 increases.
 - c. The maximum allowable increase is incorrect. The constraint $x_1^* \geq 0$ limits the increase in t to 1000.
 - d. The direction of change in ζ is incorrect. Slide 8 shows that ζ increases with t .
 - e. The objective function does change with t . Slide 8 shows that ζ increases with t .
2. Using complementary slackness (Slide 3), verify the optimality of the solution $x_1^* = 25$, $x_2^* = 75$ for the primal problem in Slide 1. What are the optimal values of the dual variables y_1^* and y_2^* , and how do they relate to the optimal objective function value?
 - a. Since $x_1^* > 0$ and $x_2^* > 0$, complementary slackness implies that the corresponding dual slack variables are zero: $y_1 + 10y_2 = 40$ and $y_1 + 50y_2 = 70$. Solving this system yields $y_1^* = 32.5$ and $y_2^* = 0.75$. Substituting these values into the dual objective function gives $100(32.5) + 4000(0.75) = 6250$, which equals the optimal primal objective function value ζ^* , confirming strong duality.
 - b. This solution violates complementary slackness since $x_1^* > 0$ and $x_2^* > 0$ imply $y_1 + 10y_2 = 40$ and $y_1 + 50y_2 = 70$.
 - c. These values are the primal optimal solutions, not the dual optimal solutions.
 - d. These values do not satisfy the dual constraints or complementary slackness.
 - e. These values do not satisfy the dual constraints or complementary slackness.
3. Explain the economic interpretation of the dual variables y_1^* and y_2^* obtained in the previous question. How are these values related to the sensitivity of the optimal objective function value to changes in the right-hand side of the constraints?
 - a. The dual variables represent the shadow prices of the resources. y_1^* is the shadow price of the first constraint (total resource 1), and y_2^* is the shadow price of the second constraint (total resource 2). A small increase of t in the right-hand side of the i -th constraint will increase the optimal objective function value by approximately y_i^*t . This is evident from Slide 10, where $\Delta\zeta^* = (\Delta b)^T y^*$.
 - b. The dual variables are not the optimal quantities of resources used. They represent the marginal value of additional resources.
 - c. The dual variables represent the marginal value (increase) of additional resources, not the marginal cost.
 - d. The dual variables are not related to production levels in this context. They represent the value of additional resources.
 - e. The dual variables are not reduced costs. Reduced costs relate to the change in the objective function when changing non-basic variables.

4. Consider the matrix representation of the linear program in Slide 4. If we partition the variable vector X into basic (X_B) and non-basic (X_N) variables, what is the significance of the reduced cost vector $\bar{C} = (B^{-1}N)^T c_B - c_N$ in determining the optimality of the current solution?
 - a. The reduced cost vector $\bar{C}$ represents the change in the objective function value per unit increase in each non-basic variable. If all elements of $\bar{C}$ are non-negative, increasing any non-basic variable will not improve the objective function, indicating that the current solution is optimal. If any element of $\bar{C}$ is negative, increasing the corresponding non-basic variable will improve the objective function, indicating that the current solution is not optimal.
 - b. This is incorrect. A negative reduced cost indicates that the objective function can be improved by increasing the corresponding non-basic variable.
 - c. This is incorrect. A zero reduced cost indicates that increasing the corresponding non-basic variable will not change the objective function value, but it doesn't guarantee optimality.
 - d. This is incorrect. A positive reduced cost indicates that increasing the corresponding non-basic variable will worsen the objective function.
 - e. This is incorrect. The reduced cost vector is crucial for determining optimality in the simplex method.

5. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
 - a. The analysis in Slides 7 and 8 shows that increasing the right-hand side of the second constraint by t leads to $x_1^* = 25 - t/40$ and $x_2^* = 75 + t/40$. Since x_1^* and x_2^* must remain non-negative, we have $25 - t/40 \geq 0$ and $75 + t/40 \geq 0$. This implies $-1000 \leq t \leq 1000$. The objective function becomes $\zeta = 6250 + 0.75t$, showing an increase in ζ with increasing t . Thus, x_1 decreases, x_2 increases, and ζ increases. The maximum allowable increase before infeasibility is $t = 1000$.
 - b. This option incorrectly states that x_1 increases and x_2 decreases. The calculations in Slide 7 show the opposite effect.
 - c. This option has the correct trend for x_1 , x_2 , and ζ , but the maximum allowable increase t is incorrectly calculated. The constraint $25 - t/40 \geq 0$ gives $t \leq 1000$.
 - d. This option incorrectly states that ζ decreases. The calculations in Slide 8 show that ζ increases with increasing t .
 - e. This option incorrectly states that ζ remains unchanged. The calculations in Slide 8 show that ζ increases with increasing t .

6. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
 - a. From Slide 7, increasing the right-hand side of the second constraint by t leads to the new optimal solution $x_1^* = 25 - \frac{t}{40}$ and $x_2^* = 75 + \frac{t}{40}$. Thus, x_1 decreases and x_2 increases. The new objective function value is $\zeta^* = 6250 + 0.75t$, showing an increase in ζ . For feasibility, we require $x_1^* \geq 0$ and $x_2^* \geq 0$, which implies $25 - \frac{t}{40} \geq 0$ and $75 + \frac{t}{40} \geq 0$. The first inequality gives $t \leq 1000$. The second inequality gives $t \geq -3000$. Therefore, the maximum allowable increase is $t \leq 1000$.
 - b. Incorrect. While the objective function increases, the changes in x_1 and x_2 are not correctly predicted. The upper bound on t is also incorrect.

- c. Incorrect. The upper bound on t is incorrect. The analysis in Slide 7 shows a tighter constraint on t .
 - d. Incorrect. The change in ζ is correctly predicted, but the changes in x_1 and x_2 are incorrect.
 - e. Incorrect. The objective function does change with t . The analysis in Slide 7 shows that ζ increases with t .
7. Using complementary slackness (Slide 3), verify the optimality of the solution $x_1^* = 25$, $x_2^* = 75$ for the primal problem in Slide 1. What are the optimal values of the dual variables y_1^* and y_2^* , and how do they relate to the optimal objective function value?
- a. Since $x_1^* > 0$ and $x_2^* > 0$, complementary slackness implies that the corresponding dual slack variables are zero: $y_1 + 10y_2 = 40$ and $y_1 + 50y_2 = 70$. Solving this system (as shown in Slide 3) yields $y_1^* = 32.5$ and $y_2^* = 0.75$. The optimal dual objective function value is $100y_1^* + 4000y_2^* = 100(32.5) + 4000(0.75) = 6250$, which equals the optimal primal objective function value ζ^* , verifying strong duality and thus the optimality of the solution.
 - b. Incorrect. These values do not satisfy the complementary slackness conditions and do not lead to the correct dual objective function value.
 - c. Incorrect. These values do not satisfy the complementary slackness conditions and do not lead to the correct dual objective function value.
 - d. Incorrect. These values are the primal optimal values, not the dual optimal values. They also do not satisfy the complementary slackness conditions.
 - e. Incorrect. These values do not satisfy the complementary slackness conditions and do not lead to the correct dual objective function value.
8. Explain the economic interpretation of the dual variables y_1^* and y_2^* obtained in the previous question. How are these values related to the sensitivity of the optimal objective function value to changes in the right-hand side of the constraints?
- a. Incorrect. While the interpretation is close, it's more accurate to say they represent shadow prices, which reflect the marginal increase in the objective function value, not necessarily profit.
 - b. The dual variables y_1^* and y_2^* represent the shadow prices of the resources associated with the first and second constraints, respectively. A shadow price indicates the marginal increase in the optimal objective function value (ζ^*) for a one-unit increase in the corresponding resource (right-hand side of the constraint). Slide 10 explicitly shows that the change in ζ^* due to a change in b is given by $(\Delta b)^T y^*$. Therefore, the sensitivity of ζ^* to changes in the right-hand side is directly given by the dual variables.
 - c. Incorrect. Shadow prices represent the marginal increase in the objective function value, not the marginal cost of resources.
 - d. Incorrect. The dual variables represent shadow prices, not the optimal quantities of resources.
 - e. Incorrect. The dual variables are not slack variables; they are shadow prices.
9. Consider the matrix representation of the linear program in Slide 4. If we partition the variable vector X into basic (X_B) and non-basic (X_N) variables, what is the significance of the reduced cost vector $\bar{C} = (B^{-1}N)^T c_B - c_N$ in determining the optimality of the current solution?
- a. The reduced cost vector $\bar{C}$ represents the change in the objective function value per unit increase in each non-basic variable. If all elements of $\bar{C}$ are non-negative ($\bar{C} \geq 0$), then increasing any non-basic variable will not improve the objective function value, indicating that the current solution

- is optimal. If any element of $\bar{C}$ is positive, it means that increasing the corresponding non-basic variable will improve the objective function, indicating that the current solution is not optimal.
- b. **Incorrect.** A positive reduced cost indicates that the solution is ***not*** optimal because increasing the corresponding non-basic variable improves the objective function.
 - c. **Incorrect.** A negative reduced cost indicates that the solution is ***not*** optimal because increasing the corresponding non-basic variable improves the objective function.
 - d. **Incorrect.** The reduced cost vector is crucial for determining optimality in the simplex method.
 - e. **Incorrect.** A zero reduced cost for a non-basic variable doesn't necessarily imply optimality; all reduced costs must be non-negative for optimality.
10. a. In the linear program presented in Slide 1, if the right-hand side of the second constraint ($10x_1 + 30x_2 \leq 4000$) is increased by t , how does this affect the optimal values of x_1 , x_2 , and ζ ? What is the maximum allowable increase t before the current optimal solution becomes infeasible?
- A. From Slide 7, increasing the right-hand side of the second constraint by t leads to the new optimal solution $x_1^* = 25 - \frac{t}{40}$ and $x_2^* = 75 + \frac{t}{40}$. Thus, x_1 decreases and x_2 increases as t increases. From Slide 8, the new optimal objective function value is $\zeta^* = 6250 + 0.75t$, indicating an increase in ζ . For the solution to remain feasible, we must have $x_1^* \geq 0$ and $x_2^* \geq 0$, which implies $25 - \frac{t}{40} \geq 0$ and $75 + \frac{t}{40} \geq 0$. The first inequality gives $t \leq 1000$, and the second inequality gives $t \geq -3000$. Therefore, the maximum allowable increase in t before the solution becomes infeasible is 1000.
 - B. **Incorrect.** While the objective function increases, the changes in x_1 and x_2 are opposite to what is stated.
 - C. **Incorrect.** The upper bound on t is calculated incorrectly.
 - D. **Incorrect.** The objective function increases, and the changes in x_1 and x_2 are opposite to what is stated.
 - E. **Incorrect.** The objective function does increase with t .
- b. Using complementary slackness (Slide 3), verify the optimality of the solution $x_1^* = 25$, $x_2^* = 75$ for the primal problem in Slide 1. What are the optimal values of the dual variables y_1^* and y_2^* , and how do they relate to the optimal objective function value?
- A. Since $x_1^* = 25 > 0$ and $x_2^* = 75 > 0$, complementary slackness implies that the corresponding dual slack variables are zero: $y_1 + 10y_2 = 40$ and $y_1 + 50y_2 = 70$. Solving this system gives $y_1^* = 32.5$ and $y_2^* = 0.75$. The optimal objective function value of the dual problem is $100y_1^* + 4000y_2^* = 100(32.5) + 4000(0.75) = 6250$, which equals ζ^* , verifying strong duality and the optimality of the primal solution.
 - B. **Incorrect.** These values are the primal optimal solution, not the dual optimal solution. Strong duality is not satisfied.
 - C. **Incorrect.** These values do not satisfy the dual constraints.
 - D. **Incorrect.** These values are the coefficients from the objective function, not the dual optimal solution. Strong duality is not satisfied.
 - E. **Incorrect.** These values are the coefficients from the objective function, switched. Strong duality is not satisfied.